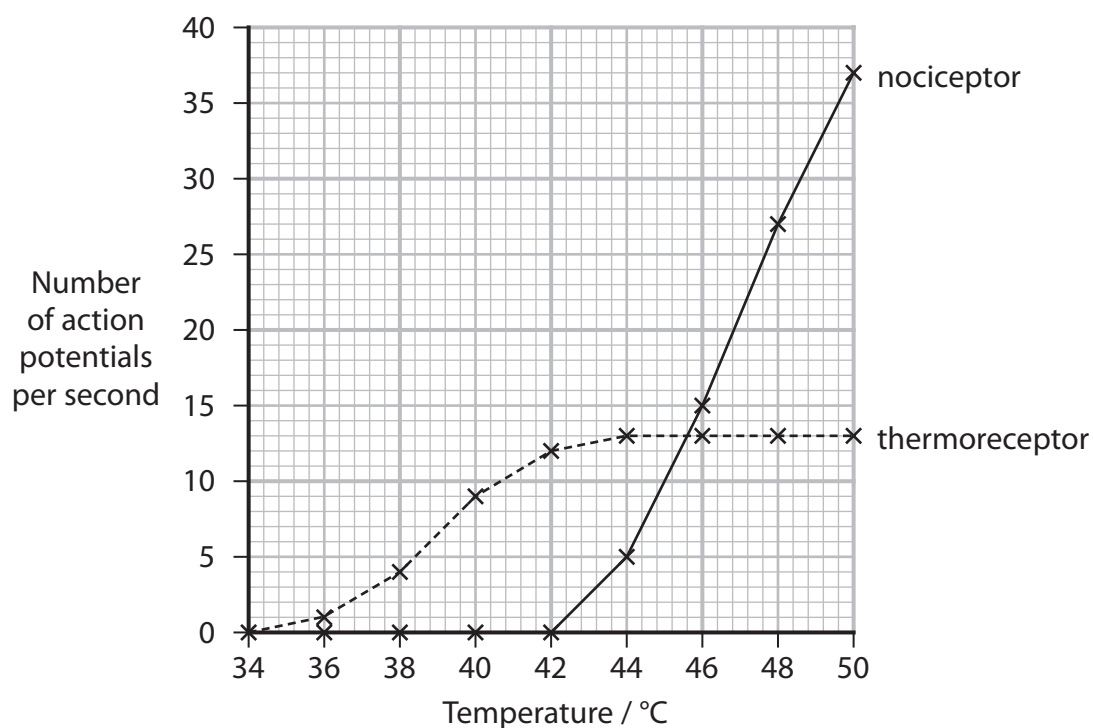


4 The nervous system of an organism enables it to respond to a stimulus.

- (a) Nociceptors and thermoreceptors are receptors that are sensitive to changes in temperature.

An investigation compared the effects of temperature on the activity of thermoreceptors and nociceptors in the skin of volunteers.

The graph shows the number of action potentials produced when each type of receptor was stimulated at different temperatures.



- (i) Which is the ratio of the action potentials produced at 48°C by nociceptors compared with thermoreceptors?

(1)

- ☐ A 0.40:1
- ☐ B 2.0:1
- ☐ C 2.07:1
- ☐ D 2.1:1

- (ii) Nociceptors are better than thermoreceptors at preventing serious damage to tissues at high temperatures.

Comment on the extent to which the results of this investigation support this statement.

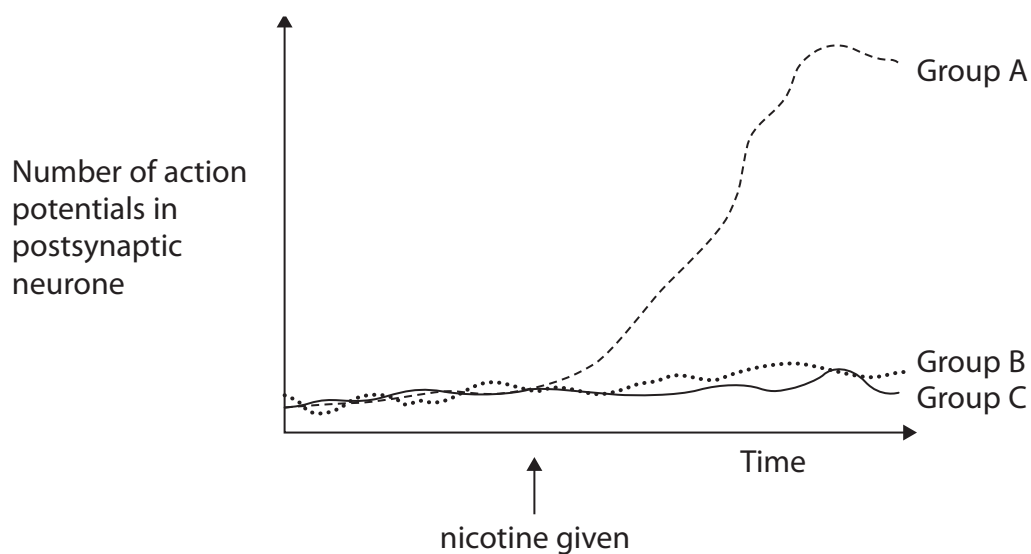
(4)

(b) Scientists used three groups of mice (A, B and C) to investigate the effect of nicotine on the transmission of nerve impulses.

- Group A were given nicotine
- Group B were not given nicotine
- Group C were given nicotine but were deficient in the gene that produces the acetylcholine receptor on the postsynaptic membrane.

The mice in groups A and C were given the same dose of nicotine (0.5 mg / kg) over the same time period.

The results are shown in the graph.



Explain the results of this investigation.

Use your own knowledge and the results in the graph to support your answer.

(4)

[illegible]

(Total for Question 4 = 9 marks)